

Related Retrieval and Lightweight-Adaptation Methods

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1 How to read this note

“Zero-shot” is not synonymous with “uses no target data.” In TS-RAG and Cross-RAG, no parameter is updated on an evaluation task, but that task’s training split is still available as a retrieval knowledge base. RAFT is a different, full-shot setting: its forecasting model is trained on the target dataset. RAF and TimeRAG augment a frozen forecaster at inference. The final three methods below adapt a foundation model or its output without a retrieval index.

Method	Date and venue	Evaluation data and settings	What is trained, retrieved, and used as backbone
Cross-RAG	arXiv v3, 15 Jul. 2026; KDD’26 MILETS Workshop, 9–13 Aug. 2026	ETTh1/2, ETTm1/2, Weather, Electricity, Exchange; chronological 6:2:2 for ETT and 7:1:2 otherwise; main $L = 512, H = 64$; $k = 15, \lambda = 0.7$.	Chronos-Bolt main and MOMENT extension. Generic pretraining updates only projectors and self/cross-attention (about 4% of parameters); backbone and predictor remain frozen. At target inference, the target training split is a KB of (X_j, Y_j) pairs; cosine search is performed directly on X . No target-task parameter update.
TS-RAG	arXiv 6 Mar. 2025; NeurIPS 2025 main track	Same seven datasets, splits, and main 512–64 setting; extended horizons 96, 192, 336, 720 are also reported.	Chronos-Bolt main and MOMENT appendix. A frozen pretrained encoder retrieves top- k context–future pairs by latent Euclidean distance; projected neighbor futures and the query representation are fused by the trainable Adaptive Retrieval Mixer (ARM). Projectors/ARM are pre-trained once; backbone and retrieval encoder are frozen and there is no target-task tuning.
RAF	arXiv 18 Nov. 2024; v2, 6 Apr. 2026	Benchmark I: Weather, Traffic, ETTh1, FRED-MD, Covid Deaths, NN5 with $L \in \{50, 75, 100, 150\}$ and $H \in \{10, 15, 20\}$. Benchmark II: Tourism-M/Q, M1-M, Uber TLC, CIF-2016 with $L \in \{10, 15, 18, 21\}$ and $H \in \{3, 4, 5\}$.	Frozen Chronos Mini/Base main; Moirai Base, TimesFM, and Lag-Llama extensions. A fixed random 80% of series forms a dataset-specific KB and 20% are test series. Approximate nearest-neighbor search retrieves a past context and its future; the normalized/aligned retrieved context and future are prepended to the query. No RAF parameter is trained.
RAFT	7 May 2025 preprint; ICML, 13–19 Jul. 2025	Four ETT, Electricity, Exchange, Illness, Solar, Traffic, Weather; multivariate; $H \in \{96, 192, 336, 720\}$ (Illness: $\{24, 36, 48, 60\}$); validation-selected L , usually 720.	No foundation-model backbone. A shallow multiresolution MLP is trained per dataset. From the target training series, stride-one keys retrieve their following future patches by Pearson correlation; top- m futures are softmax-aggregated at periods 1, 2, and 4, concatenated with the query, and mapped to the forecast.

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TimeRAG	21 Dec. 2024 preprint; ICASSP 2025, 6–11 Apr. 2025	M4, covering yearly through hourly frequencies.	A historical-sequence KB is clustered/indexed; DTW selects reference inputs. References and query are rendered as text for the LLM-based forecaster. The RAG path is inference-time augmentation; the paper reports a 2.97% mean improvement over its original model.

2 Cross-RAG and TS-RAG in detail

Both papers first sample 50 million points from the Chronos pretraining corpus, use 5 million to construct a multi-domain pretraining KB, and form about 26 million training pairs and 2.8 million retrieval pairs. The downstream evaluation training splits are excluded from this *pretraining* KB. At zero-shot evaluation, however, each target training split constructs the target KB and its test split supplies queries. These two facts are compatible: the target KB changes at inference but the learned parameters do not.

TS-RAG retrieves

$$(X_j, e_j, Y_j), \quad j \in \text{TopK}_{\min} \|e(X) - e(X_j)\|_2,$$

then projects the retrieved futures and applies ARM self-attention jointly with the backbone query representation. Cross-RAG instead retrieves in the data space,

$$j \in \text{TopK}_{\max} \frac{\langle X, X_j \rangle}{\|X\| \|X_j\|},$$

projects both X_j and Y_j , uses the query against retrieved inputs as cross-attention keys and retrieved outputs as values, and also retains a query-independent retrieval self-attention path. Its final representation is

$$z = \lambda \tilde{c} + (1 - \lambda) \tilde{s}, \quad \lambda = 0.7.$$

Cross-RAG trains its added modules for 10,000 AdamW steps on the generic pretraining pairs. The published comparison uses $k = 15$ for Cross-RAG and the best $k = 5$ for TS-RAG.

The main Cross-RAG table reports Chronos-Bolt MSEs of 0.341, 0.243, 0.290, 0.143, 0.144, 0.112, and 0.064 over the seven datasets (unweighted mean 0.191), versus its reproduced TS-RAG values 0.357, 0.246, 0.298, 0.147, 0.151, 0.112, and 0.071 (mean 0.197). The reduction computed from the rounded mean column is $(0.197 - 0.191)/0.197 = 3.05\%$; recomputing the means from the seven printed cells gives 3.26%. The mean of the seven separate relative reductions is instead 3.66%. These are distinct summaries and should be named explicitly. The paper states that baseline entries combine reproductions using official implementations with results taken from the original TS-RAG paper; they are not all newly trained from scratch.

3 Fine-tuning and lightweight adaptation

Method	Date and venue	Evaluation data and settings	Training and backbone
TimesFM-ICF	31 Oct. 2024 preprint; ICML, 13–19 Jul. 2025	18 unseen Monash datasets plus ETTh1/2 and ETTm1/2; ETT horizons 96 and 192; up to 50 related examples; base maximum target history 512.	Starts from a 200M-parameter decoder-only TimesFM pretrained on about 400B points. Continued pretraining teaches separator-aware, cross-example attention. At target inference there are no gradients: related series are concatenated as demonstrations. Target-specific controls fully fine-tune TimesFM or update only its input/output MLP blocks.

Method	Date and venue	Evaluation data and settings	Training and backbone
ELF	ICML, 13–19 Jul. 2025	ETTh1/2, ETTm1/2, US-Weather, Weather, Solar, ECL, Traffic; rolling deployment; horizons 30, 96, 336; update every 200 new points.	Frozen Chronos-Tiny, TTM-r2, TimesFM, Moirai-Small, or VisionTS. A Fourier linear ELF-Forecaster is updated online using Woodbury sufficient statistics; an exponential fast/slow ELF-Weighter combines it with the frozen FM. There is no retrieval.
Beyond LoRA	17 Sep. 2024; NeurIPS 2024 Workshop on Time Series in the Age of Large Models	eICU sepsis: MeanBP and heart-rate forecasting; 4,020 samples from 1,442 patients; patient-disjoint 8:1:1 split; six-hour context sampled every five minutes ($L = 72$), three-hour target ($H = 36$).	Chronos Tiny/Small/Base main, with Mini/Large appendix. Compares full fine-tuning and LoRA with BitFit, LayerNorm tuning, VeRA, and FourierFT. Original weights remain frozen for additive PEFT; selective PEFT updates only chosen native parameters. Full-FT and LoRA comparison values are quoted from the preceding study; the four new PEFT variants are run by this paper.

The comparison provenance matters. TimesFM-ICF borrows official Monash baseline values and most non-TimesFM ETT values from the cited benchmark papers, while running TimesFM-Base and TimesFM-ICF itself. RAFT states that it validation-tunes both RAFT and all nine supervised baselines and averages three runs. These numbers therefore answer different questions and should not be combined into a single cross-paper ranking.

4 Relation to this project

Our retrieved signals (X_j, Y_j, N_j) resemble the input–output memories in the RAG methods, but our primary question is narrower: can cheap, explicitly validated mixtures or gates improve the same frozen vanilla forecast V ? The directly comparable external experiment is therefore a fresh execution of released Cross-RAG/TS-RAG code on our fixed T_3 dates, not a transcription of their 512–64 table. Their pretrained adaptation modules and our fitted ridge/gate models also use different training regimes, which must remain visible in any results table.

5 Primary sources and released code

- Cross-RAG: [paper](#); [code](#).
- TS-RAG paper, code, processed datasets, retrieval database, and checkpoints: [project release](#).
- RAFT paper and code: [paper](#); [code](#).
- RAF: [paper](#).
- TimesFM-ICF: [paper](#).
- ELF: [paper](#).
- Beyond LoRA: [paper](#).